

FinTrust Customer & Transaction Data Analysis

Week 2 — Understand & Execute | Data Analytics Track

Intern: Miancy (Mercy Chepkemai Koech)

A. Applying the Week 1 Plan

1. Originally planned: Week 1's plan called for data preparation and exploratory analysis in Week 2 (using Excel/Python), followed by KPI computation, ahead of Power BI dashboard-building in Week 3.

2. Implemented: All of this was accelerated into Week 2 itself — Excel data-quality assessment, SQL business analysis (8 questions), Python EDA with 5 visualizations, and a working Power BI dashboard with KPIs, 6 visual categories, and slicers were all completed this week.

3. What changed: The Week 1 plan treated dashboard-building as a Week 3 activity; it was brought forward into Week 2 because the brief's Week 2 deliverables explicitly required a working .pbix file.

4. Why: Building the dashboard alongside the SQL/Python analysis (rather than after) made it possible to cross-validate results across all three tools in the same week — every KPI and chart in Power BI was checked against the equivalent SQL query and Python output before being finalized.

B. Resources Used

- FinTrust_Customer_Data.csv and FinTrust_Transaction_Data.csv — imported into SQL Server (SSMS) as dbo.FinTrust_Customer_Data and dbo.FinTrust_Transaction_Data, and read directly via Python (pyodbc/pandas) and Excel.
- FinTrust_Data_Dictionary.xlsx — used as a reference for field definitions throughout, kept open separately rather than imported as a table.
- Limitation encountered: SQL Server's Import Wizard auto-typed the Risk_Review_Flag and International_Transaction fields as bit (1/0) rather than text (Yes/No) as they appear in the raw CSV — this required adjusting all relevant SQL queries and Python code to compare against 1/0 instead of 'Yes'/'No'.

C & D. Tools Used and Practical Outputs Produced

Tool	Output Produced	Filename
Microsoft Excel	Data Quality & Cleaning Workbook with live formulas (COUNTBLANK, COUNTIF, MIN/MAX/AVERAGE) plus a cleaned Transaction_Data sheet	FinTrust_Week2_DataQuality_Cleaning_Workbook.xlsx
SQL Server (SSMS)	8 business questions answered with queries, results, and interpretations	FinTrust_Week2_SQL_Analysis.sql
Python (Pandas/Seaborn, via VS Code + Jupyter, connected live to SQL Server)	Exploratory Data Analysis notebook with 5 visualizations and insight write-ups	FinTrust_Week2_Data_Analysis.ipynb

Tool	Output Produced	Filename
Power BI	Interactive management dashboard: 6 KPI cards, 6 required visual categories, slicers, and a Key Highlights panel	FinTrust_Week2_Analytics_Dashboard.pbix

E. Testing / Evaluation Log

Work Completed	Test/Evaluation	Finding	Action/Improvement	Result
SQL risk queries (Risk_Review_Flag comparisons)	Ran query comparing field to 'Yes'	Conversion error: field stored as bit, not text	Rewrote all affected queries to compare against 1 instead of 'Yes'	Queries ran cleanly; results cross-validated against Python
Power BI KPI cards	Manually recalculated all 6 KPIs in Python against raw CSV data	All 6 values matched exactly (customers, transactions, value, average, success rate, risk rate)	None needed — confirmed accuracy	Dashboard KPIs verified correct
Segment risk-rate finding (Q6)	Compared raw flagged counts vs. rate-normalized results	Raw counts misleadingly suggested Everyday was riskiest; normalized rates showed all segments within ~1 point of each other	Rewrote Q6 to calculate rate instead of raw count	More defensible, accurate finding used in Business Findings
Monthly transaction value trend (Power BI line chart)	Recalculated monthly totals directly from CSV in Python	Jan ■188.5M, Feb ■175.8M, Mar ■196.2M — matched the dashboard's dip-then-rise shape exactly	None needed — confirmed accuracy	Trend chart verified correct, not a rendering artifact

F. Important Decisions Documented

Decision: Compare Risk_Review_Flag and International_Transaction to 1/0 instead of 'Yes'/'No' in all SQL and Python code. **Why:** SQL Server's Import Wizard auto-converted these text fields to bit type on import. **Evidence:** Direct query error ("Conversion failed when converting the varchar value 'Yes' to data type bit") and confirmation via DISTINCT query. **Effect:** All 3 affected SQL queries and the Python risk-rate chart were corrected before finalizing results.

Decision: Impute missing Device_Type and Location values in Transaction_Data as 'Unknown' in a separate Cleaned_Transaction_Data sheet, rather than deleting rows or altering the raw data. **Why:** Only 96 rows (0.8%) were affected per field — deleting them would lose otherwise valid transaction data.

Evidence: COUNTBLANK formulas in the Excel workbook. **Effect:** Raw data preserved for audit; a usable cleaned version exists for any downstream analysis needing complete categorical fields.

Decision: Normalize Q6 (customer segment risk) by transaction volume (rate) rather than reporting raw flagged counts. **Why:** Raw counts are misleading when comparing groups of very different sizes (Everyday has 3-4x the transactions of other segments). **Evidence:** Side-by-side comparison showing raw counts and normalized rates told different stories. **Effect:** The normalized version was used in the final Business Findings instead of the raw-count version.

G. Limitations Identified

- **Data-quality:** 96 missing values each in Device_Type and Location (0.8% of Transaction_Data) — imputed as 'Unknown' rather than resolved with certainty.

- **Methodological:** Risk_Review_Flag is a synthetic educational label, not a real fraud/risk determination — findings about "risk" describe patterns in this label, not verified financial risk.
- **Technical:** SQL Server's automatic bit-type conversion of Yes/No fields required query adjustments; anyone re-running this analysis against the raw CSV directly (e.g. in SQLite) would need to reverse this adjustment back to comparing against text values.
- **Scope:** The monthly trend analysis only covers 3 months (Jan-Mar 2026) — not enough data to confirm whether the Feb dip is a real seasonal pattern or a one-off fluctuation.
- **Assumption:** Analysis treats all synthetic data as representative of real FinTrust operations for learning purposes, per the assignment's data disclaimer.

Part E — Business Findings

Finding 1: International transactions carry nearly double the risk-flag rate of domestic transactions

Evidence: SQL analysis shows international transactions are flagged at 36.88% (177 of 480 transactions), compared to 18.88% for domestic transactions (2,175 of 11,520) — confirmed visually in the Power BI dashboard.

Business Meaning: This is the strongest, most actionable risk signal in the dataset. FinTrust should weight risk-review resources more heavily toward international transactions and prioritize international status as a key input if a formal risk-scoring model is developed in later weeks.

Finding 2: Customer segment is not a meaningful risk differentiator once normalized by volume

Evidence: Raw flagged counts made Everyday customers look riskiest (1,089 flagged transactions), but normalizing by transaction volume shows all four segments sit within a tight 19.29%-20.33% risk-rate band — Premium is actually marginally highest, not Everyday.

Business Meaning: Segment-based risk targeting would be a poor strategy; risk drivers lie elsewhere (see Finding 1). This also demonstrates the importance of rate-based rather than raw-count analysis when comparing groups of different sizes.

Finding 3: Nearly 8% of all transactions fail or reverse, representing a real operational friction point

Evidence: 5.25% of transactions are Failed and 2.72% are Reversed - a combined 956 transactions out of 12,000 - while 90.47% succeed.

Business Meaning: An 8% failure/reversal rate is high enough to plausibly drive customer support volume and erode trust in the platform. This deserves a focused Week 3 investigation into whether failures cluster around specific channels or transaction types.

Finding 4: The Everyday segment is the core of FinTrust's business, not Premium

Evidence: Everyday customers make up nearly half the customer base (711 of 1,500) and generate the most total transaction value (■258.7M) - more than the other three segments combined.

Business Meaning: FinTrust's growth and retention strategy should center on the Everyday segment rather than treating it as lower priority relative to Premium. Losing Everyday customers would have a disproportionately large revenue impact.

Finding 5: Mobile App is the dominant channel, making platform reliability there a high-stakes priority

Evidence: Mobile App accounts for 5,102 of 12,000 transactions (42.5%) - more than double the next-highest channel (POS, 2,393) - while carrying an average transaction value comparable to other channels (■47,115).

Business Meaning: Any downtime, bugs, or UX friction on the Mobile App channel would affect the largest share of FinTrust's transaction volume. Reliability investment should be prioritized here over lower-traffic channels like USSD, even though USSD serves an important financial-inclusion role worth preserving.